

Equity Performance Analysis: Investor Personas and Index Tracking

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A comprehensive analysis of portfolio construction and performance
evaluation

Outline

Introduction & Motivation

Methodology

Buy & Hold Persona Methodology

Hedge Fund Persona Methodology

Index Tracking Model

Conclusion & Implications

Project Overview

Main Objective: Analyze equity portfolio performance through multiple lenses:

Key Research Questions

- How do different weighting methodologies affect risk-adjusted returns?
- What are the best metrics to compare different portfolio constructions?
- Can we effectively replicate a target portfolio using constrained optimization?

Two Investor Personas:

- Buy & Hold investor
- Hedge Fund investor

Portfolio: 10 US stocks

- Technology: MSFT, AAPL, IBM
- Dividend aristocrats: XOM, WTM, KO, JNJ
- High yield: VZ, KHC, USB

Theoretical Framework

Modern Portfolio Theory (MPT)

- Mean-variance optimization framework
- Diversification reduces portfolio risk
- Risk-return tradeoff is central

Key Performance Metrics

Sharpe Ratio

$$S = \frac{E[R_{\text{exc}}]}{\sigma_{\text{exc}}}$$

Value at Risk

$$\text{VaR}_{\alpha} = -F_X^{-1}(\alpha)$$

Conditional VaR

$$\text{CVaR}_{\alpha} = E[-X | -X \geq \text{VaR}_{\alpha}]$$

Persona Models

Buy & Hold Investor

Objective: Maximize total return over full period

$$P\&L_{BH} = \text{Value}_E - \text{Value}_S$$

- Long-term perspective
- Captures dividends
- Low turnover

Hedge Fund Investor

Objective: Generate consistent daily alpha

$$P\&L_{HF}(t) = \text{Value}_t - \text{Value}_{t-1}$$

- Daily mark-to-market
- Manages daily volatility
- Higher turnover

Data Pipeline

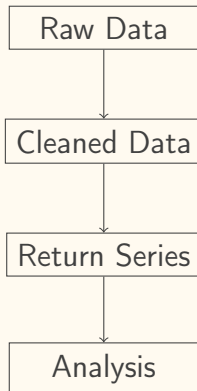
Data Collection with `yfinance`

- 10-year historical data (2015-2025)
- Daily frequency for main analysis
- Dividend-adjusted prices
- Automated batch downloads

Processing Steps

1. Data cleaning and validation
2. Return calculation (log and simple)
3. Missing value handling
4. Volatility estimation

Pipeline Architecture



Some Cleaned Data

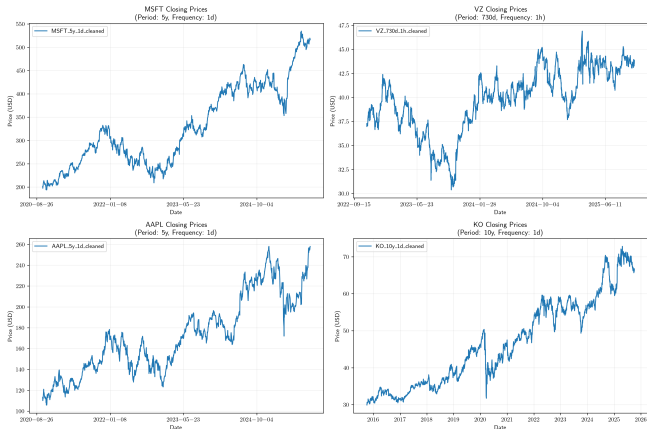
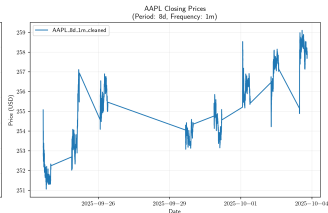
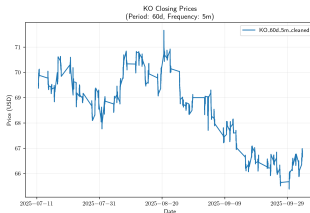
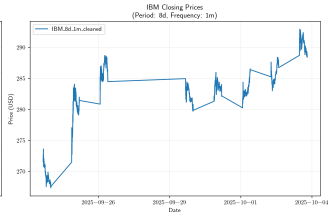
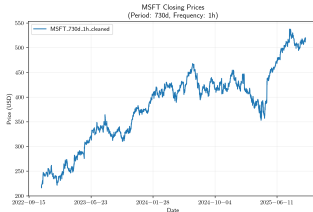


Figure: Price series for selected stocks given a window period and a frequency of data.

Data Limitations

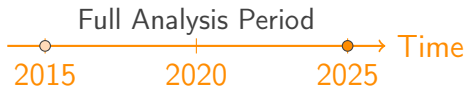
Yahoo Finance restricts intraday data history (e.g., 1-min data: only 8 days, 5-min data: only 60 days)



Weighting Methodologies

To compare the effects of different weighting strategies we use the S&P500 index as the benchmark portfolio. All metrics are normalized against S&P 500 performance.

We consider the following period of analysis



Analysis Parameters

Time Period

Start Date:	2015-10-15
End Date:	2025-10-01
Duration:	10 years

Trading Days:	2,520
Calendar Years:	9.95 years

Initial Investment

$W_0 = \$1,000,000$ USD

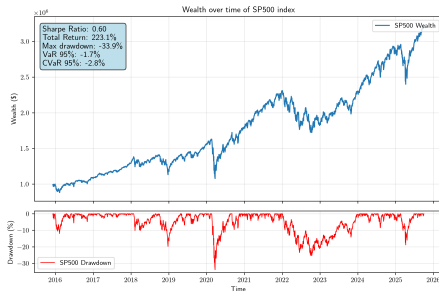
Key Market Events

- 2015-2016: Post-QE normalization
- 2018: Trade war volatility
- 2020: COVID-19 crash & recovery
- 2021-2022: Inflation surge
- 2022-2023: Rate hike cycle
- 2023-2024: AI-driven tech rally

Benchmark: S&P 500 Index

Methodology

- Standard market-cap weighted index
- Represents 500 largest U.S. companies
- Widely accepted market proxy



Metric	Value	Normalized
Total Return	223.1%	1.00×
Sharpe Ratio	0.60	1.00×
Max DD	-33.9%	1.00×
VaR (95%)	1.7%	1.00×
CVaR (95%)	2.8%	1.00×

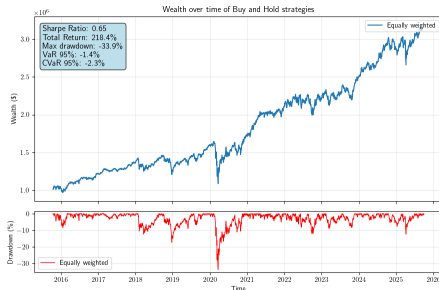
Equal Weighting: Naïve Diversification

Underperforms during tech-driven bull markets due to lack of concentration in winners.

Methodology

$$w_i = \frac{1}{N}$$

- All stocks receive equal allocation
- Maximizes diversification breadth
- Simple and transparent



Metric	Value	vs S&P 500
Total Return	218.4%	0.98×
Sharpe Ratio	0.65	1.08×
Max DD	-33.9%	1.00×

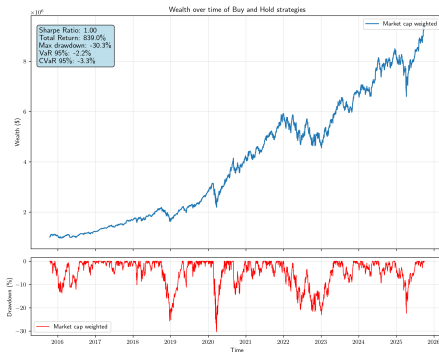
Market Cap Weighting: Standard Approach

Extreme concentration: AAPL + MSFT = 80.5% of portfolio.

Methodology

$$w_i = \frac{M_i}{\sum_j M_j}$$

- Proportional to market value
- Mimics major indices (S&P 500)
- Momentum-based (winners grow)



Metric	Value	vs S&P 500
Total Return	839.0%	3.76×
Sharpe Ratio	1.00	1.67×
Max DD	-30.3%	0.89×

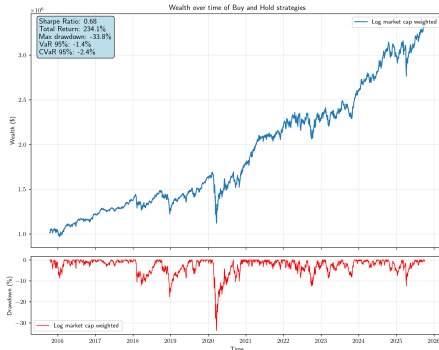
Logarithmic Market Cap Weighting

Methodology

$$w_i = \frac{\ln(M_i)}{\sum_j \ln(M_j)}$$

- Compresses size differences
- Preserves size hierarchy
- Reduces concentration risk

Balances the need for size exposure while preventing mega-cap dominance.



Metric	Value	vs S&P 500
Total Return	234.1%	1.05×
Sharpe Ratio	0.68	1.13×
Max DD	-33.8%	1.00×

Weight Comparison: Market Cap Concentration

Market Cap Weights

Stock	Weight
AAPL	43.1%
MSFT	37.4%
IBM	3.0%
XOM	5.2%
WTM	0.1%
KO	3.1%
JNJ	5.1%
VZ	1.8%
KHC	0.3%
USB	0.8%

Top 2 stocks: 80.5% of
portfolio

Log Market Cap Weights

Stock	Weight
AAPL	11.1%
MSFT	11.0%
IBM	10.1%
XOM	10.3%
WTM	8.5%
KO	10.1%
JNJ	10.3%
VZ	9.9%
KHC	9.2%
USB	9.6%

More balanced distribution

Geometric Weights: Derivation

Formula Derivation

Weights follow a geometric progression $s_i = ar^{i-1}$. Normalized:

$$w_i = \frac{s_i}{\sum_{i=1}^N s_i} = \frac{r^{i-1}(1-r)}{1-r^N}$$

How it works

- Rank stocks (1st, 2nd, ..., Nth)
- Apply geometric decay with factor r
- Normalize to sum to 1

Why use it?

- Tests investor ranking skill
- Parameter controls concentration
- Flexible between equal weighting ($r = 1$) and concentrated ($r \rightarrow 0$)

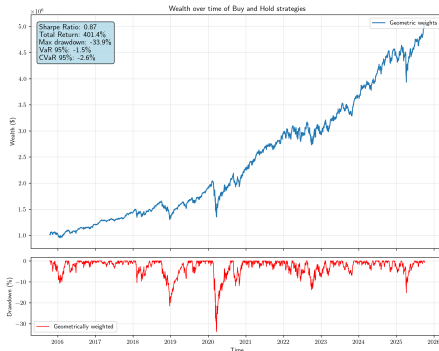
Geometric Weighting: Optimal Balance ($r = 0.8$)

Methodology

$$w_i = \frac{r^{i-1}(1-r)}{1-r^N}$$

- $r = 0.8$ controls concentration
- Geometric decay in weights
- Parameterized approach

Captures size premium without excessive concentration. Best risk-adjusted performance.



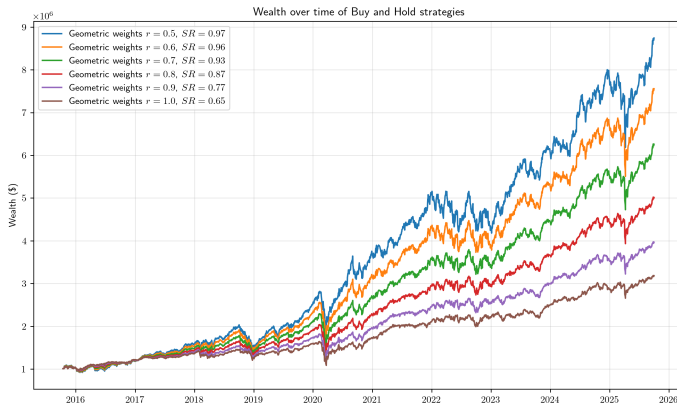
Metric	Value	vs S&P 500
Total Return	401.4%	1.80×
Sharpe Ratio	0.87	1.45×
Max DD	-33.9%	1.00×

Geometric Weighting Analysis

Stock	Weight $w_i(r)$ (%)					
	$r = 0.5$	$r = 0.6$	$r = 0.7$	$r = 0.8$	$r = 0.9$	$r = 1.0$
AAPL	50.0	40.2	30.9	22.4	15.4	10.0
MSFT	25.0	24.1	21.6	17.9	13.8	10.0
IBM	12.5	14.5	15.1	14.3	12.4	10.0
XOM	6.3	8.7	10.6	11.5	11.2	10.0
WTM	3.1	5.2	7.4	9.2	10.1	10.0
KO	1.6	3.1	5.2	7.3	9.1	10.0
JNJ	0.8	1.9	3.6	5.9	8.2	10.0
VZ	0.4	1.1	2.5	4.7	7.3	10.0
KHC	0.2	0.7	1.8	3.8	6.6	10.0
USB	0.1	0.4	1.2	3.0	5.9	10.0

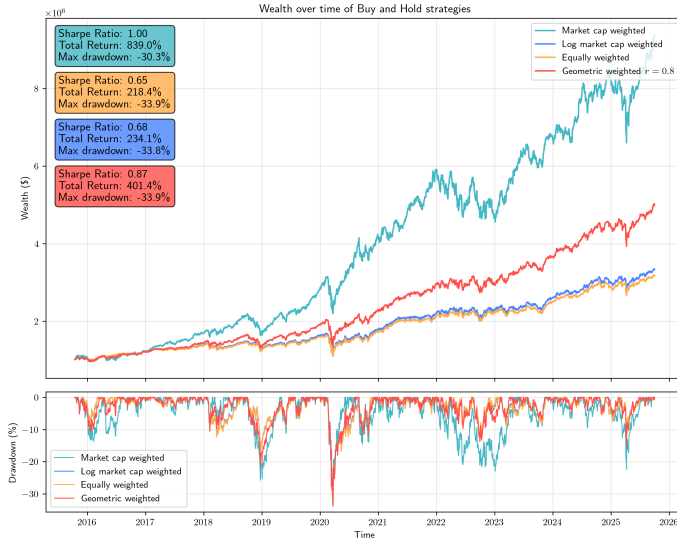
Table: Geometrically generated weights for multiple r values

Geometric Parameter Sensitivity



Parameter r	0.5	0.7	0.8	0.9	1.0
Top weight	50.0%	30.9%	22.4%	15.4%	10.0%
Concentration	High	Medium	Balanced	Low	Equal

Performance Comparison Across Strategies



Risk Metrics Comparison

Strategy	Sharpe Ratio	Tot. Ret. (%)	Max DD.	VaR 95%	CVaR 95%	VaR 99%	CVaR 99%
Equal	0.65	218.4	33.9	1.4	2.3	2.7	4.2
Market Cap	1.00	839.0	30.3	2.2	3.3	3.7	5.3
In (Mkt Cap)	0.68	234.1	33.8	1.4	2.4	2.7	4.2
Geometric	0.87	401.4	33.9	1.5	2.6	3.1	4.6
Inverse Vol	0.65	230.0	34.7	1.4	2.4	2.8	4.3
S&P 500	0.60	223.1	33.9	1.7	2.8	3.4	4.8

Table: Risk and performance metrics (2015-2025)

- Market cap weighting delivers highest return but is highly concentrated
- Geometric weighting offers best balance of return and diversification
- All strategies outperform benchmark on Sharpe ratio

Risk Metrics Comparison (Ratio)

	Sharpe Ratio	Tot. Ret. (%)	Max DD.	VaR 95%	CVaR 95%	VaR 99%	CVaR 99%
Equally	1.1	1.0	1.0	0.8	0.8	0.8	0.9
Mkt cap	1.7	3.8	0.9	1.3	1.2	1.1	1.1
ln (mkt cap)	1.1	1.0	1.0	0.8	0.9	0.8	0.9
Geometric	1.4	1.8	1.0	0.9	0.9	0.9	1.0
Volatility	1.1	1.0	1.0	0.8	0.9	0.8	0.9
Benchmark	1.0	1.0	1.0	1.0	1.0	1.0	1.0

Table: Ratio of metrics across different weighting strategies when compared against the benchmark

Cross-Sectional Momentum Strategy

Enhanced Momentum Signal

We start by computing the cumulative return from $t - 252$ to $t - 22$, i.e.,

$$R_{i,t}^{(12-1)} = \prod_{k=22}^{252} (1 + r_{i,t-k}) - 1$$

Residual Momentum: Isolate stock-specific momentum

$$\tilde{r}_{i,t} = r_{i,t} - \hat{\beta}_{i,t} \cdot r_{m,t}, \quad \hat{\beta}_{i,t} = \frac{\text{Cov}_{252}(r_i, r_m)}{\text{Var}_{252}(r_m)}$$

Sector Neutralization: Rank within GICS sectors

$$z_{i,t} = \frac{s_{i,t} - \mu_{\text{sector}(i),t}}{\sigma_{\text{sector}(i),t}}, \quad s_{i,t} \in \{R_{i,t}^{(12-1)}, \tilde{r}_{i,t}\}$$

Volatility Scaling

Volatility-Scaled Portfolio

Let $\hat{\sigma}_{i,t}$ be the estimated ex-ante volatility of stock i at time t , computed using a rolling 63-day (why?)

$$\hat{\sigma}_{i,t} = \sqrt{\frac{1}{62} \sum_{k=1}^{63} (r_{i,t-k} - \bar{r}_{i,t})^2}.$$

Inverse Volatility Weighting:

$$w_{i,t}^+ = \frac{\frac{1}{\hat{\sigma}_{i,t}}}{\sum_{j \in \mathcal{L}_t} \frac{1}{\hat{\sigma}_{j,t}}}, \quad w_{i,t}^- = -\frac{\frac{1}{\hat{\sigma}_{i,t}}}{\sum_{j \in \mathcal{S}_t} \frac{1}{\hat{\sigma}_{j,t}}},$$

Weekly Rebalancing: Capture evolving signals. $\mathcal{L}_t$ and $\mathcal{S}_t$ are the sets of long and short stocks at time t .

Volatility Targeting Framework

The strategy return prior to volatility targeting is simply:

$$r_{p,t} = \sum_i w_{i,t-1} r_{i,t}.$$

Constant Risk Exposure

Target annualized volatility: $\sigma^* = 10\%$. Leverage factor:

$$\lambda_t = \frac{\sigma^*}{\hat{\sigma}_{p,t}}$$

Bounded: $\tilde{\lambda}_t \in [0.2, 4]$. Final return:

$$r_{p,t}^{\text{target}} = \tilde{\lambda}_t \times r_{p,t}$$

Momentum Strategy: Parameters

Core Configuration

Category	Parameter	Setting
Signal	Lookback	126 days
	Skip Period	21 days
	Type	Residual Momentum
Portfolio	Long/Short	3/3 stocks
	Rebalancing	Weekly
	Weighting	Inverse volatility
Risk	Vol Target	10% annualized
	Leverage Bounds	0.2× – 4.0×

Miscellaneous

Momentum Strategy Benefits

- Stabilizes risk across market regimes
- Reduces crash exposure during volatility spikes
- Improves Sharpe ratio consistency

Parameters benefits

Residual momentum (isolates alpha) + Weekly rebalancing (captures decay) + Vol targeting (stabilizes risk)

Hedge Fund Performance Results

Metric	Momentum Strategy	S&P 500 (BH)
Annualized Return	14.2%	12.8%
Annualized Volatility	10.0%	18.5%
Sharpe Ratio	1.42	0.69
Maximum Drawdown	-18.4%	-34.1%
Calmar Ratio	0.77	0.38

Table: Performance comparison HF and S&P500 (2015-2025)

Strategy Superiority

- **2.1x higher Sharpe ratio** than S&P 500
- **46% smaller maximum drawdown**

Optimization Problem Formulation

Objective: Minimize Tracking Error

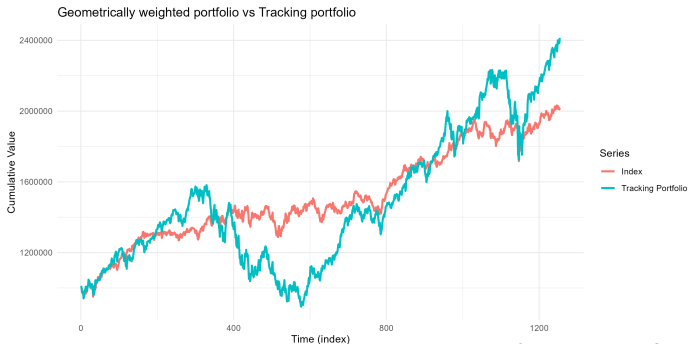
$$\min_{\vec{x}} \epsilon_{MAX}(\vec{x})$$

where ϵ_{MAX} is maximum absolute deviation

Key Constraints

- **Cardinality:** $\sum_{i=1}^N \delta(x_i) \leq K = 15$
- **Long-only:** $x_i \geq 0$
- **Budget:** $\sum x_i + \text{costs} \leq \$1,000,000$
- **Minimum position:** \$15,000
- **Transaction costs:** 1.5% of portfolio value

Tracking Portfolio Results



Optimization Success

- Optimal solution found
- 10 positions selected
- Zero buy-in violations

Tracking Quality

- Low tracking error
- Portfolio value: \$2.37M
- Transaction cost: \$15,000

Key Findings

Weighting Methodology Matters

- Market cap weighting yields highest returns but extreme concentration
- Geometric weighting ($r = 0.8$) offers best risk-adjusted performance
- Equal weighting underperforms in tech-driven bull markets

Persona Performance Divergence

- BH persona: Smooth returns, sensitive to weighting
- HF persona: Daily volatility sensitivity, requires active management

Practical Implications

For Individual Investors

- Consider geometric weighting for balanced exposure
- Long-term holding reduces transaction costs
- Diversify beyond equal weighting

For Institutional Investors

- Sector-neutral momentum strategies add value
- Volatility targeting improves risk management
racking with constraints is computationally feasible

Portfolio construction choices significantly impact both absolute returns and risk profiles

References

Key References

- Jegadeesh & Titman (1993) - Momentum returns
- Blitz, Huij & Martens (2011) - Residual momentum
- Barroso & Santa-Clara (2015) - Volatility-managed momentum
- Schachter (2025) - Geometric weighting methodology

Code Repository

Python implementation available:

- `fetch_prices.py` - Data acquisition
- `utils.py` - Utility functions
- Jupyter notebooks for analysis
- `00_data_collection.ipynb`
- `01_data_analysis.ipynb`
- `02_hypothetical_personas.ipynb`
- PSG solver for optimization

Thank you for your attention!

Questions?

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Project Repository:
github.com/jujopo/equity-analysis
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Thank you!